BROADER ECONOMY IN TRANSITION · BASE SCENARIOS (SSP/RCP)

What are the environmental consequences of micro- and meso-level decisions?
designing, operating, deploying or adopting technologies, in diverse contexts and over the transition

which units operate?

how they operate?

Pillar I · Market consequences
Which suppliers respond, when and where, with what environmental profile

Pillar II · Process consequences
How processes and their environmental flows respond to design, operation and context

Common strategy
local markets and processes as instances of broader ones; simplifications tested

PhD I — Task 1.1
Building a scenario-consistent framework for marginal suppliers, from model perturbation to heuristics

PhD II — Task 2.1
Wrapping process models of any sophistication in an interoperable interface, instantiated by scenario and decision

PhD I — Task 1.1
Testing marginal-supplier heuristics against structural model responses

Energy
Pillar I case

PhD III — Task 2.2
AI-assisted model building, evaluated where the answer is known

MSc I — Task 1.2
Testing heuristics against past metal capacity, then deriving marginal supply from dynamic MFA

Metallurgy
shared case

MSc II — Task 2.3
Calibrating engineering models against reported plant performance, and testing what generic datasets miss

MSc III — Task 1.3
Disaggregating scenario regions to the local markets where decisions play out

Fertilizer
shared case

PhD III — Task 2.2
Representing an emerging process before deployment, as alternative configurations

FROM KNOWN TO UNKNOWN

Develop and evaluate against a reference model

Confront with observed and reported data

Extend where no full reference exists

OUTCOMES

Scenario-consistent marginal suppliers
and the errors of common heuristics, characterized

Cross-case synthesis
joint assessment of market and process effects

Interoperable process-model classes
and an evaluated workflow to build more

TIMING
Years of activity, 1 to 5
1 2 3 4 5

Consistent, discriminating prospective consequential LCA for environmental decisions across contexts
under which conditions representation choices change the decisions supported, in the cases studied